# Introduction to Statistics — Quiz #2 (60 minutes)

**October 20, 2026 (Tuesday)**

Section: _____ Cadet Number: _____ Name: _____ Score:

- All solutions must include a detailed step-by-step explanation.
- If an answer has more than four decimal places, round to the **fourth decimal place**.
- **Reference table** is provided on the last page of the exam.

1. A multiple-choice test has $10$ questions, and each question has $5$ choices. A student picks every answer at random, so the probability of getting each question right is $1/5$. Let $X$ be the number of questions the student gets right out of the $n = 10$ questions. Answer the following questions. [30 points]

(1) Find the probability mass function $f(x)$ of $X$.

(2) Find the probability that the student gets at least one question right, $P(X \geq 1)$.

(3) Find the expectation and variance of the random variable $X$.

2. A robot shoots at a target. Each shot scores $1$, $2$, $3$, or $4$ points, and each of these four scores has probability $1/4$. The scores of different shots are independent. Answer the following questions. [30 points]

(1) Let $X_1$ and $X_2$ be the scores of the first and second shots. Find the probability mass function $f(\bar{x})$ of the average score $\bar{X} = (X_1 + X_2)/2$. (Fill in the table below.)

| $\bar{x}$ | | | | | | | |
|---|---|---|---|---|---|---|---|
| $f(\bar{x})$ | | | | | | | |

(2) Find $E(\bar{X})$ and $\mathrm{Var}(\bar{X})$.

3. The lifetime of portable flashlights made at a factory has mean $\mu = 2{,}000$ hours and standard deviation $\sigma = 50$ hours. Let $X_1, X_2, \ldots, X_n$ be the lifetimes (in hours) of $n = 100$ flashlights randomly selected from this factory. Find the probability that their sample mean $\bar{X}$ is not greater than $1{,}990$ hours, $P(\bar{X} \leq 1{,}990)$. [20 points]

4. The time a bus takes to reach its last stop follows a normal distribution with standard deviation $\sigma = 1$ minute. A random sample of $n = 17$ buses is taken and their travel times are recorded. Find the probability that the sample variance $S^2$ is greater than 2, $P(S^2 > 2)$. [20 points]

**Reference Table** For $Z \sim N(0,1)$ and $V \sim \chi^2(n)$, let $z_\alpha$ and $\chi^2_\alpha(n)$ be the values satisfying $P(Z > z_\alpha) = \alpha$ and $P(V > \chi^2_\alpha(n)) = \alpha$.

| | | | |
|---|---|---|---|
| $z_{0.4207} = 0.2$ | $z_{0.3446} = 0.4$ | $z_{0.3085} = 0.5$ | $z_{0.1587} = 1$ |
| $z_{0.0668} = 1.5$ | $z_{0.0228} = 2$ | $z_{0.0062} = 2.5$ | $z_{0.0013} = 3$ |
| $\chi^2_{0.99}(16) = 5.81$ | $\chi^2_{0.05}(16) = 26.30$ | $\chi^2_{0.025}(16) = 28.85$ | $\chi^2_{0.01}(16) = 32.00$ |
| $\chi^2_{0.99}(17) = 6.41$ | $\chi^2_{0.05}(17) = 27.59$ | $\chi^2_{0.025}(17) = 30.19$ | $\chi^2_{0.01}(17) = 33.41$ |